AI in China

From AI investment to adoption

China's AI push is shifting from ambition to execution. Investment support, and limited infrastructure and regulatory constraints, as well as early signals of adoption, are driving varied economic effects on growth, FX and credit.

What China aims to achieve: China's AI strategy reflects a national drive for technological leadership, with policymakers targeting AI adoption across more than 100 industries and a 90% penetration rate by 2030. The strategy is supported by open-model economics and a rapid buildout of public and private AI infrastructure. While macro evidence remains fragmented, we see signs of AI adoption, led by manufacturing and ICT, especially among larger corporates.

Sustainability: no major constraints: Sustainability-related factors are set to support, not constrain, China's AI expansion. Adequate power, fuel supply, permitting and spare capacity should enable further AI infrastructure expansion, while water stress remains a localized issue. On adoption, policy priorities are shifting toward workforce reskilling and AI content governance. By contrast, our thematic analysis of the US suggests a shift from AI experimentation toward execution and ROI, alongside a convergence of supply-side constraints (eg, power, labor, permitting, and local opposition).

Credit: commercialisation and funding: At the sector level, China TMTs are not only investing but beginning to show clearer commercial benefits, most directly reported by Alibaba, Baidu, Tencent, Kuaishou and Lenovo. Meanwhile, more funding is needed for infrastructure-led platforms, accordingly to our liquidity walks, and could become more complex as compute demand expands and as Chinese TMT companies extend beyond domestic procurement into overseas data centres and GPU capacity.

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Macroeconomics: AI's macro impact comes in several channels at different horizons. While the near-term contribution is likely to be led by exports and AI infrastructure capex, we think the longer-term contribution to growth will come through higher total factor productivity, as firms use AI to improve efficiency across manufacturing and services. Our other observations are that the most inflationary phase of China's AI-related infrastructure buildout is likely behind us and AI has yet to create enough jobs, with demand insufficient to counter the property downturn and automation.

FX & rates: Given the limited near-term growth impact of AI-driven productivity, we see FX and rates being driven more by the AI export cycle than by productivity gains from AI adoption. China's AI boom is more supportive of the CNY than it is bearish for CGBs. For rates, AI should keep the PBoC accommodative and CGB yields relatively low. The stronger growth impulse is more relevant for CNY NDIRS, where it reduces easing expectations and leaves the 5y sector more exposed than the policy-anchored front end.

Understanding China's AI roadmap

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China's state-led AI push aims for broad economy-wide adoption, but progress reporting remains uneven. To help investors gauge the pace of deployment, we map reported adoption trends against key industry indicators.

China's AI roadmap: Clear targets and wide industry coverage

China's official promotion of artificial intelligence dates back to its first policy in 2016 linking AI with Internet+ [1]. In March 2024, AI+ development was included in government annual budget report for the first time. And in August 2025, China furthered the AI adoption details with release of the "AI+ Action Plans", an economy-wide AI adoption program, with explicit penetration targets set for "new intelligent terminals and agents – 70% by 2027 then to 90% by 2030. Six priorities are also targeted, including **science & technology, industry, consumption, livelihood, governance and global cooperation**. The plan was formally incorporated into the national development plans of the 15th Five-Year Plan (2026-2030), in March 2026.

Following the national plans, numerous government agencies (eg, MIIT, MOFCOM, NEA) have published opinions on each of the priorities or sub-areas. For example, for industry and consumption alone, we have found **nine industry groups and >100 targeted sectors or business activities (see Appendix)**. Notably, not all the sectors come with numerical targets and many are mentioned in policy documents as case studies and potential scenarios.

AI models: improving openness and cost advantages

While China does not explicitly discourage closed-source models, the term **"open model"** appears nine times in the **AI+ Action Plan**, particularly under the **AI + Global Cooperation** pillar. By contrast, references to **closed-source models** are largely absent from both national and industry-level policy documents.

Coupled with the government's promotion of open models, we also observe growing "openness" in the latest model launches across a number of developers. Since 2024, many developers have shifted to **Apache 2.0** or **MIT licenses**, which allow weight download, fine-tuning, commercial usage, modification and self-service.

[1] "Internet+ and Artificial Intelligence Three-Year Action and Implementation Plan", National Development and Reform Commission, Ministry of Science and Technology, Ministry of Industry and Information Technology, Cyberspace Administration of China, May 2016

A clear inflection point came with DeepSeek R1, which combined frontier-level performance with fully open weights under an MIT license. Shortly thereafter, Moonshot AI made a dramatic shift in strategy, releasing Kimi K2 under a modified MIT license[2] in July 2025 and moving away from its previous API-centric approach. Baidu, after years of advocating closed-source development, also pivoted by open-sourcing the entire ERNIE 4.5 model family under Apache 2.0 in early 2025.

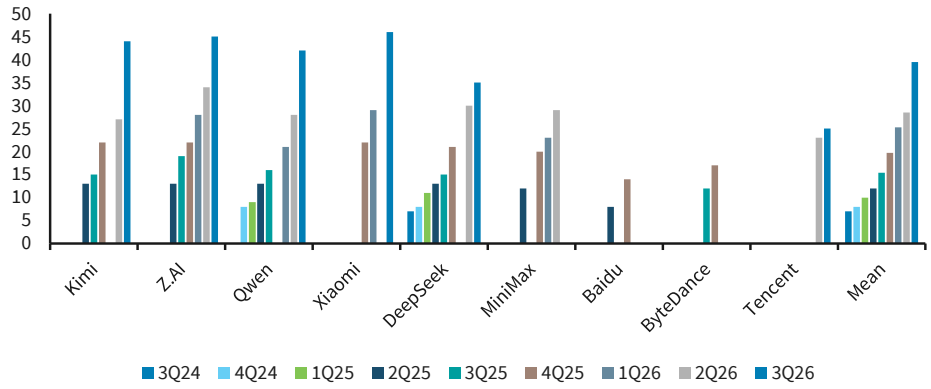
FIGURE 1. More "open" licensing for Chinese models

Developer	2023	2024	2025	2026
Deepseek	Open weights	Open weights	MIT	MIT
Kimi	Closed	Closed	Modified MIT	Modified MIT
Alibaba	Custom	Mixed	Apache	Apache
Zhipu AI	Open	Open	MIT	MIT
Baidu	Closed	Closed	Apache	Apache
Tencent	Closed	Community	Community	Apache

Source: Company data, Barclays Research

Chinese models continued to make rapid gains in performance alongside greater openness. Artificial Analysis data (Figure 2) shows that models released in 3Q2026 posted an average intelligence score of 40, led by Xiaomi's MiMo-V2.6-Pro (46) and Z.AI's GLM-5.3 (45). This marks a sharp improvement from the average score of 29 for models released in 3Q2025 and 15 for those launched a year earlier.

FIGURE 2. Intelligence of Chinese models continued to improve

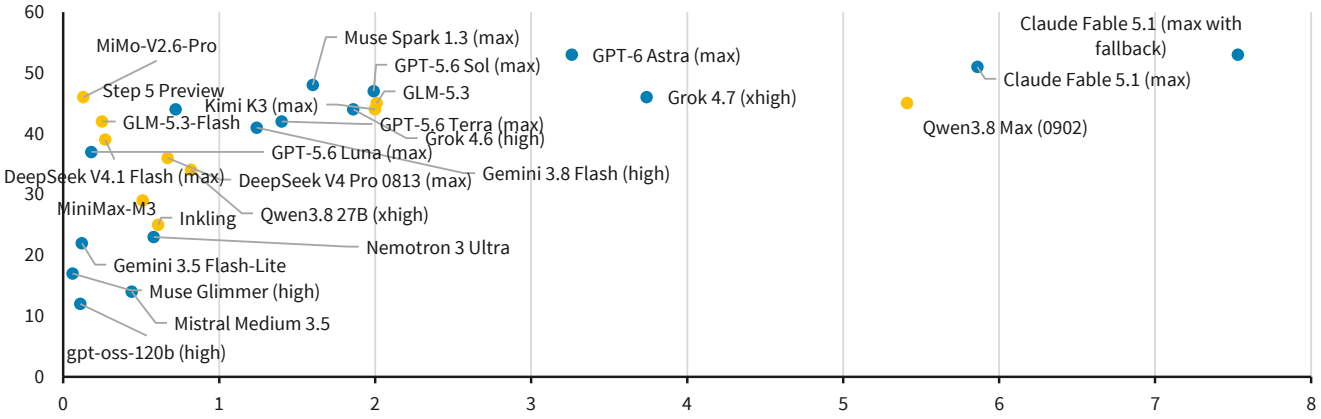


Source: Artificial Analysis, Barclays Research

While leading US frontier models still score higher on intelligence benchmarks, with Claude Fable 5.1 reaching 53, many Chinese models (shown in amber) offer comparable performance at significantly lower cost (Figure 3). This cost advantage could accelerate AI adoption across China.

[2] Kimi's modified MIT: If a company or product exceeds certain scale thresholds (reported as **more than 100 million monthly active users or more than US$20 million in monthly revenue**), it must prominently display attribution to Kimi in its product. In simple terms, large companies cannot use the model invisibly; they must acknowledge that Kimi technology is being used.

FIGURE 3. **Intelligence Index vs. Cost per task**



Amber = Chinese models Y-axis = Intelligence Index (higher, better); X-axis = USD per task (lower, better)
Source: Artificial Analysis, Barclays Research

Industry size: The CNY10trn AI target

The China Academy of Information and Communications Technology (CAICT) quantified the scale of core AI industry at **CNY1.2trn** (or USD174 billion) in 2025, recording 40% y/y growth.

And by 2030, the target expects a near 10-fold rise to **CNY10trn** by 2030 [3], or a 53% CAGR over 2026-2030. Note that this figure can be understood as value-added revenue without specifying the type of AI (see the Macroeconomic section for the detailed analysis and macro implications).

Capital: The CNY2trn data centre investment to support compute capacity

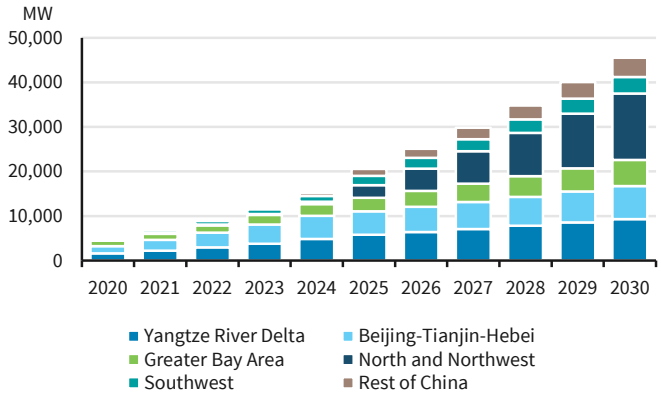
The second indicator is Beijing's push to ramp up data-centre investment over the next five years. As the physical backbone of AI computing, data centres offer investors a clear gauge of computing capacity, capital deployment and infrastructure readiness. According to press reports, China is drafting a **CNY2trn** nationwide data-centre investment program. The plan could also mandate that **80% of procurement be sourced domestically**, including AI chips[4].

It is unclear if this amount will be solely allocated to data centres, but a simple estimate assuming a cost of CNY40mn/MW suggests another **50GW of data centre capacity is set to be built**. This supports some of the market forecasts such as BNEF's 45.5GW operating capacity by 2030 (Figure 4).

[3] "China's AI-related industries to top 10 trln yuan by 2030: official", the Supreme People's Procuratorate of the PRC, 6 March 2026

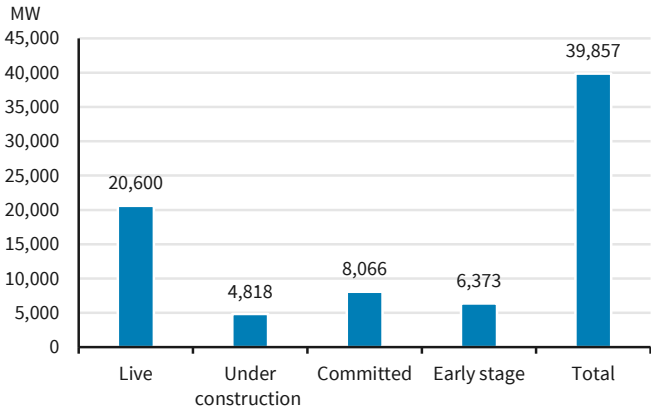
[4] "China prepares two trillion yuan plan to fund nationwide AI build-out", 9 June 2026, the Business Times

FIGURE 4. China's data centre capacity may more than double reaching 45GW in data by 2030



Source: BNEF, Barclays Research

FIGURE 5. Status of data centre capacities



Source: BNEF, Barclays Research

Adoption rate: Manufacturing and ICT adoption likely ahead

While clear government targets, the explicit goal of reaching 70% penetration by 2027 and 90%penetration by 2030 are not clearly defined given the lack of details regarding "penetration", "agent" and "intelligent terminals".

Official data show penetration surpassed 30% among manufacturers with annual revenue above CNY20mn by end-2025[5]. For non-manufacturing, earlier data indicate adoption rates of 13% in retail, 10% in telecommunications and 7% in healthcare[6]. Gartner estimates just 5% of Chinese enterprises had AI in production in 2025, but expects the figure to rise sharply to 50% by 2027[7].

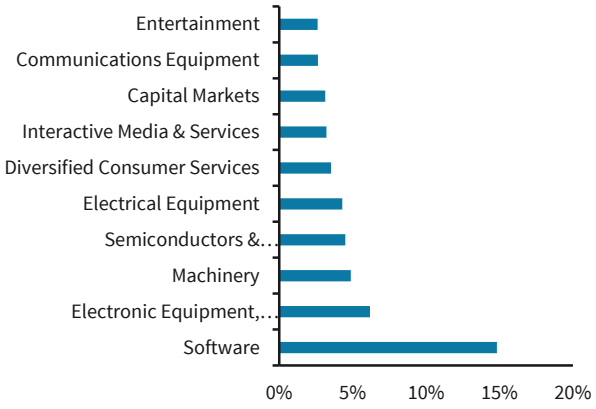
To gain further insight on the adoption rates, our analysis identified over 500 large Chinese companies with nearly 17,000 mentions of AI in filings or public disclosures, led by software and manufacturing sub-sectors (Figure 6). When grouped according to the nine broad sectors under the China's AI+ Action Plans (Figure 7), **manufacturing** dominates with a 62% share, ahead of ICT (19%) and commercial services (14%). In comparison, based on the same exercise tracking US companies' filings, the finding points to similar dominance by the **software** sector, but followed by many services sectors such as **financial, IT and professional services (Figure 8).**

[5] "China's core AI industry scale tops 1.2 trln yuan in 2025: official", Xinhua, 3 March 2026

[6] "China's AI computing power development is booming, and it is becoming a new driving force for the development of the digital economy", CCTVC, 14 December 2023

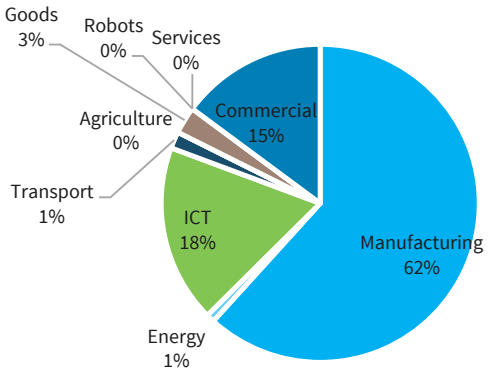
[7] "Global Companies to Adopt Chinese AI at 50% Rate by 2027", the Chosun Daily, 9 August 2026

FIGURE 6. Chinese companies mentioning AI are largely software and manufacturing…



Contribution of AI mentioned in total company filings
Source: AlphaSense, Barclays Research

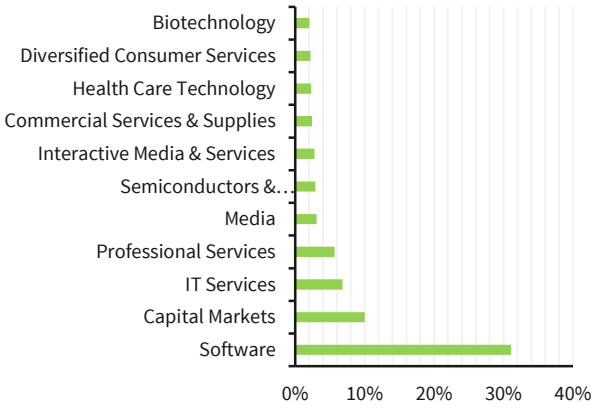
FIGURE 7. … in the format of China's AI+ classifications, manufacturing and ICT lead in adoption



Contribution of AI mentioned in company filings
Source: AlphaSense, Barclays Research

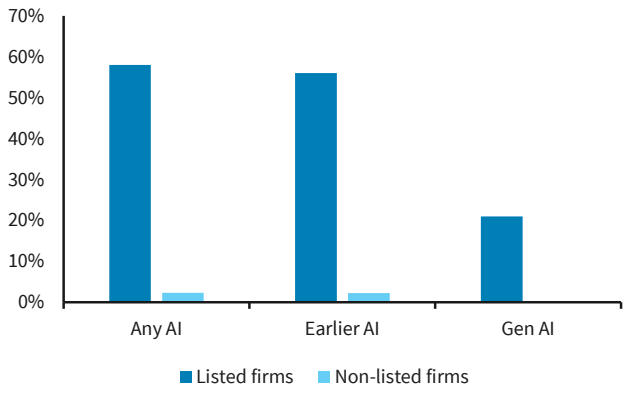
Investors should note that this data set represents only a fraction of China's more than 55mn registered private enterprises[8], suggesting AI adoption remains skewed toward larger firms. A recent World Bank and Peking University study, based on job-posting analysis, found AI penetration reached 58% among listed companies and 21% for generative AI, compared with just 2.3% and 0.2%, respectively, among non-listed firms[9].

FIGURE 8. Unlike China, US companies mentioning AI are more software and services than manufacturing



Contribution of AI mentioned in company filings
Source: AlphaSense, Barclays Research

FIGURE 9. Adoption rates: listed vs non-listed Chinese firms



Source: Peking University, World Bank, Barclays Research

AI application development and consumer adoption

Additionally, investors can also track innovation momentum, with GenAI models and applications seeing a rising share in global AI-related patents, from 4.2% in 2017 to 8.7% in 2025 (Figure 10), driven particularly by LLMs, diffusion models and multi-modal systems[10]. China-based inventors published over 43,000 GenAI patent families in 2024 and 2025. And as of August

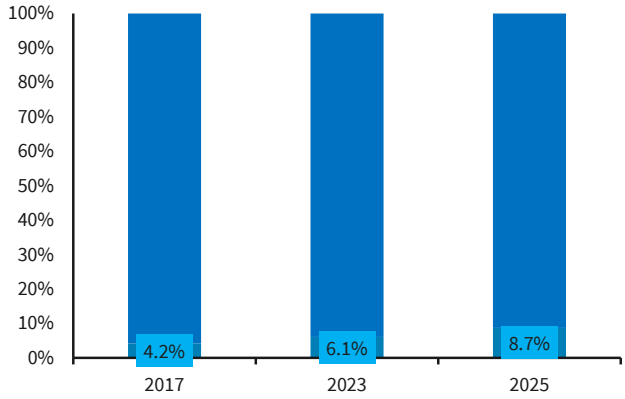
[8] "Overwhelming majority of business entities in China from private sector", Xinhua News, 6 November 2024

[9] "Now Hiring: Firm-Level Adoption of Generative AI in China"Li, Liu, Wang, Wang and YuNational School of Development, Peking University, World Bank, July 29, 2026

[10] "Patent Trends Update in GenAI", World Intellectual Property Organization, 2026

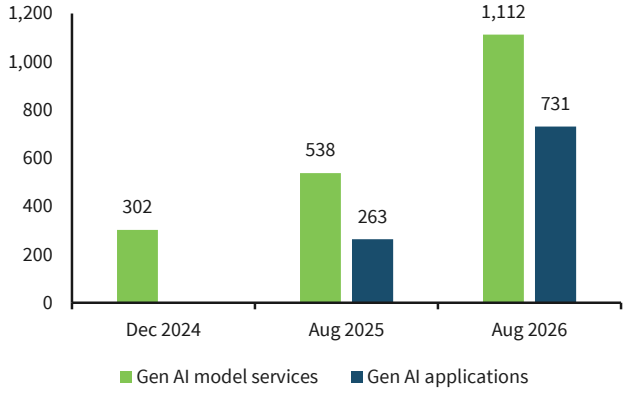
2026, total registered AI model services[11] already reached 1,112, double the year-earlier level, while the number of registered GenAI applications rose to 731[12] (Figure 11).

FIGURE 10. GenAI's growing share in global AI-related patents



Source: World Intellectual Property Organization, Barclays Research

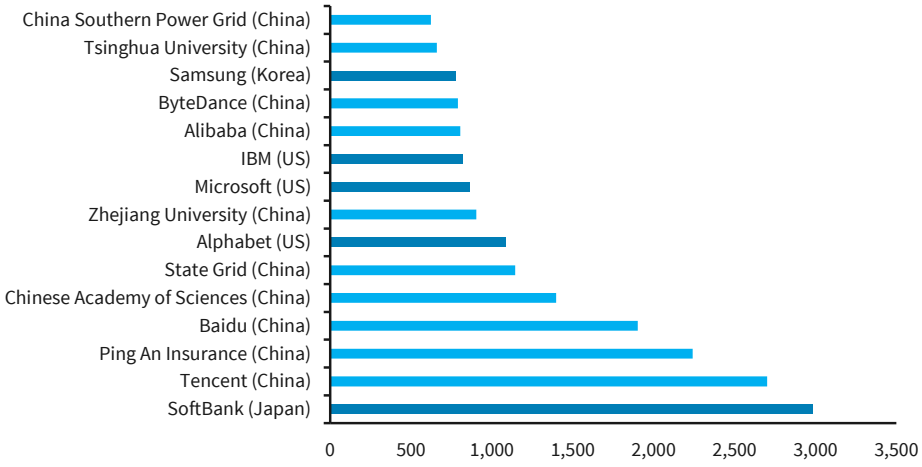
FIGURE 11. Growing development of GenAI products



Source: Cyberspace Administration of China, Barclays Research

In terms of GenAI patent owners, **10 of the world's top 15 companies** are headquartered in China, led by **Tencent, Ping An Insurance and Baidu**. Outside China, Japan's SoftBank Group is the world's largest, followed by Alphabet (7th), Microsoft (9th), IBM (10th) and Samsung Electronics (13th).

FIGURE 12. GenAI patents during 2014-2025: Chinese companies dominate



Source: World Intellectual Property Organization, Barclays Research

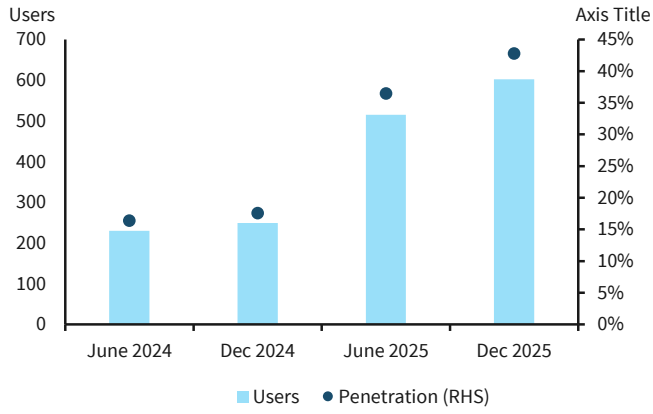
Lastly, while China's AI ambition clearly prioritizes industry adoption, we consider consumer-level AI adoption equally important, as final consumer goods and services (such as AI PC and humanoids) will depend heavily on end-users' evolving AI acceptance and dependence. According to China Internet Network Information Center (CNNIC), total GenAI users reached 602mn people as of December 2025 (249mn in December 2024 and 515mn in June 2025),

[11] Refers to underlying AI technology, model training and hosting, API interfaces, or cloud computing platforms provided by technology vendors. Examples: OpenAI API, Alibaba Cloud Bailian, Google Cloud Vertex AI, etc.

[12] Refers to AI software with an independent product form factor and a complete user interface (UI), specifically designed to meet specific needs. Examples: ChatGPT, Kimi, Midjourney (standalone drawing software).

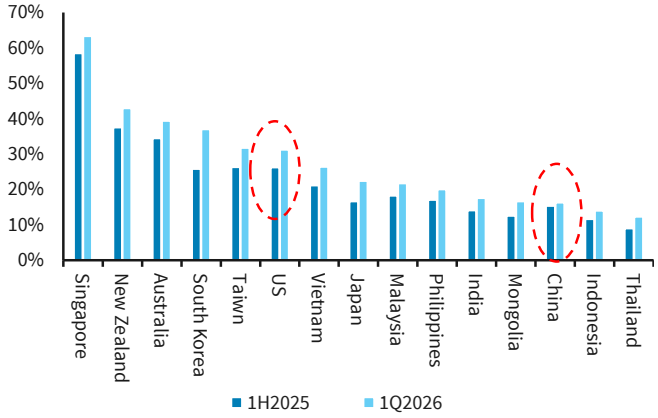
covering 42.8% of population (Figure 13). But this differs from the calculation by Visual Capitalist[13] that covers the working population, which suggests China's 16.4% penetration rate is lower than the US and many Asian economies (Figure 13).

FIGURE 13. Generative AI users in China



Source: CNNIC, Barclays Research

FIGURE 14. AI adoption rate of working population



Source: Visual Capitalist, Barclays Research

Sustainable Investing: Power, water, labour and regulations

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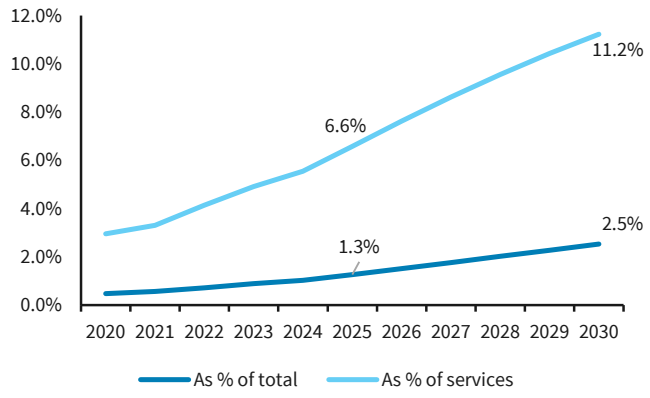
From a sustainability perspective, we examine critical constraints on AI deployment, including power and water availability, social consequences, and regulatory priorities. Several of these challenges differ from those in the US (see Thematic section), despite similarly rapid AI adoption.

Power supply: Unlikely a binding constraint

Data centres are energy intensive and water dependent. According to BNEF, data centres consumed 131 TWh of electricity in 2025 (1.3% of total consumption), and that could expand to 329 TWh (2.5%) by 2030 (Figure 15), assuming 5% of annual power demand growth during 2026-30. In a more optimistic cases, Wood Mackenzie and the National Electricity Administration forecast 774 TWh and 800 TWh, respectively, by 2030, lifting their share of power consumption to 6.0% and 6.2%, respectively (Figure 16).

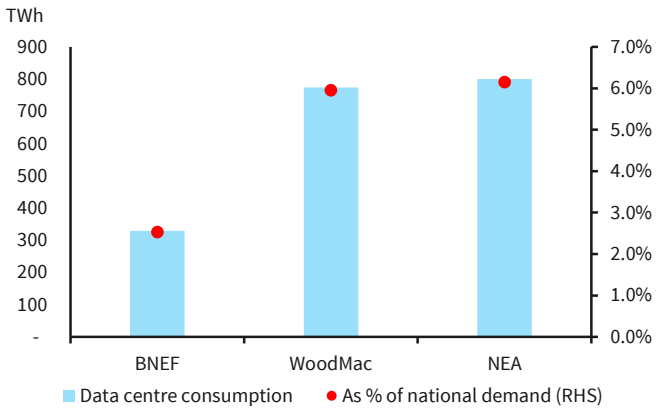
[13] Source: Microsoft AI Diffusion Report Q1 2026. Based on estimates of Microsoft users that use AI at least 90 minutes per month, adjusted for the working-age population (Age 15-64).

FIGURE 15. Data centres: rising electricity share but still at low single digits by 2030...



Source: National Energy Administration, BNEF, company data, Barclays Research

FIGURE 16. ...alternative forecasts point to 6% of power demand by 2030

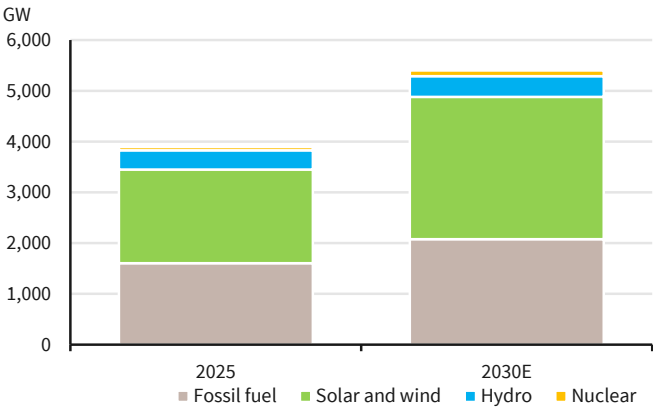


Source: Wood Mackenzie, National Electricity Administration, Barclays Research

To examine's China's electricity supply security and capacity buffers, the country is first fully self-sufficient in thermal coal, which was used to generate 54% of power generation in 2025. Renewables, hydro, and nuclear accounted for 22%, 14% and 5%, respectively, which are largely dependent on domestic technologies with limited imported materials. Oil and gas, which still needs to be imported, accounted for only 3-4% of power generation.

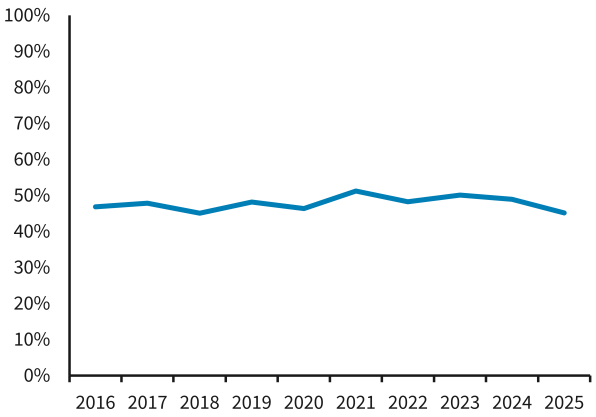
On additional supplies, the 15th Five-Year Plan (2026-2030) published in early 2026 included a simultaneous expansion of non-fossil and coal capacities (Figure 17). This is an important playbook as China's coal-fired buildout usually takes only two to three years with a less complex permitting process, which could play a key supporting role in the case of higher-than-expected data centre power demand. Lastly, China's current load factor of coal-fired power has been under-utilised, leaving headroom to support unexpected demand (Figure 18). The plan also includes larger investment in transmission capex, which should support more transmitted power from Western to Eastern China, where computing demand is larger (Weekly Spotlight: China's power grid ambition, 2 February 2026).

FIGURE 17. China's electricity security is not an issue given self-sufficiency in coal supply



Source: National Energy Administration, Barclays Research

FIGURE 18. Coal power load factor has trended down



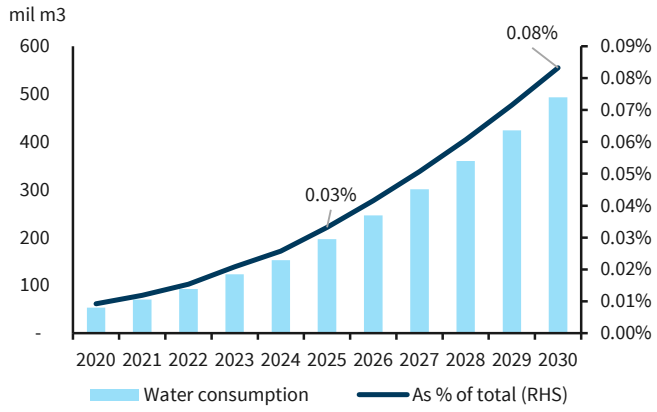
Huaneng Power as a proxy
Source: Company data, Barclays Research

Water risk: More local than systemic

Water remains a highly localised constraint. We estimate data-centre power consumption will require approximately 194mn m^3 and 493mn m^3 of water in 2025 and 2030, respectively, based on a water usage effectiveness (WUE) of 1.5[14] . Although this accounts for only 0.03% and 0.08% of national water supply (Figure 19), significantly below many Asian data-centre markets, local water stress may still emerge where demand is concentrated and supply is constrained.

As discussed in China TMTs: Green efforts with room to strengthen amid digital boom (26 November 2024), under China's East Data West Computing (EDWC) initiative, launched in 2022, 10 high-density, low-carbon data-centre clusters are being developed across eight hubs. Although western provinces offer ample power supply, water availability remains a growing constraint in certain regions, particularly Inner Mongolia and Ningxia (Figure 20).

FIGURE 19. Data centres: water consumption likely to remain 0.1% of national water supplies by 2030...



Source: National Energy Administration, BNEF, company data, Barclays Research

FIGURE 20. ... but water stress risks could expand with more data centres built in western China

Provinces	Location	Water stress (2030 BAU)
Inner Mongolia	North	Extremely high
Ningxia	West	Extremely high
Hebei	North	Extremely high
Chengdu	West	High
Beijing	North	High
Tianjin	North	High
Zhejiang	East	High
Shanghai	East	High
Guizhou	West	Medium-high
Gansu	West	Low
Chongqing	West	Low
Jiangsu	East	Low
Guangdong	South	Low

The key cluster bases of data centres under the EDWC program
Source: World Resources Institute, Barclays Research

In addition to data centres, we note that China is also building semiconductor production capacity, adding extra pressure to energy and water supplies. As discussed in Asia semiconductors: revisiting today's disruptions and tomorrow's challenges (16 September 2026), we calculate that from 2025 to 2030 the broader tech sector's share of national electricity demand should expand by 120bp and its share of water consumption should rise by 13bp, broadly matching data centres' shares. We also highlighted that semiconductor Chinese companies Semiconductor Manufacturing International Corp (SMIC) and ChangXin Memory Technology (CXMT) may face water stress challenges given their local exposure.

Labor disruption: legal complexity in focus

On the social side, China's AI strategy aims to maximise productivity gains while preserving labour-market stability, with the AI+ Action Plans placing equal emphasis on employment risks and job creation.

[14] Water usage effectiveness = total water consumed (Liters) / total IT equipment energy donsumed (kWh); lower score means more efficient.

On one hand, China's strong AI prospects could, over time, open doors to productivity gains and new occupations. Based on Barclays' AI Automation and Application (3A) Framework (AI in action: The market for skills, 22 September 2026), only around 20% of core workplace skills are highly replicable by current AI capabilities. AI's productivity benefits can extend across much of the labour market, suggesting that productivity-enhancing effects could prove more widespread than direct automation effects. The decade of the robot could also create new forms of demand for human expertise.

Near-term disruption, however, may be difficult to quantify. China's strong social safeguards and policy focus on employment stability could mask the extent of labour displacement, making it harder for investors to gauge the effects of AI adoption on the workforce.

The legal case in Hangzhou

In April 2026, the Intermediate People's Court of Hangzhou ruled against an AI company that had replaced a quality-assurance supervisor with LLMs and subsequently terminated his employment[15]. The court found that:

- Replacing workers with AI does not qualify as a "major change in objective circumstances" under China's Labour Contract Law and therefore does not, by itself, justify redundancy-related dismissals.
- Offering a reassigned role with a 40% pay cut was unreasonable and amounted to transferring the costs of AI-driven transformation to employees.

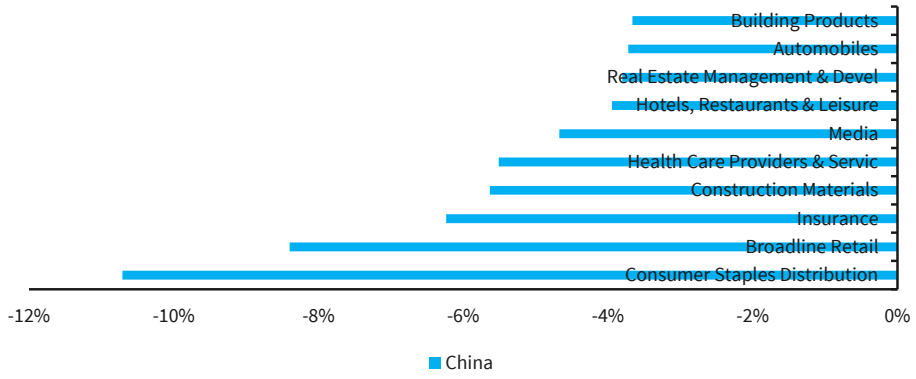
The case underscores China's willingness to protect workers affected by AI adoption, highlighting how social-stability objectives may influence the pace and visibility of labour-market disruption.

Signals from the details of employment changes at large companies

In the absence of explicit disclosure, workforce data may provide one of the clearest indicators of AI-related labour disruptions. Companies reporting sizeable headcount reductions warrant closer scrutiny, in our view. At the sector level, the largest average workforce declines in China were recorded in consumer staples distribution, retail, insurance, construction materials, healthcare providers, media, hotels & leisure, real estate, automobiles and building products (Figure 21).

[15] "Chinese court defends labor rights in new AI-replacement case", the State Council Information Office of the PRC, 30 April 2026

FIGURE 21. 10 industries with largest labor force reduction in China



Changes from 2024 to 2025; public listed companies reporting employee data
Source: Bloomberg, Barclays Research

Our review of company-level AI strategies and workforce disclosures suggests AI adoption is becoming increasingly embedded in production, operational and IT processes. **However, none of the companies examined explicitly attributed changes in workforce size to AI deployment** (Figure 22).

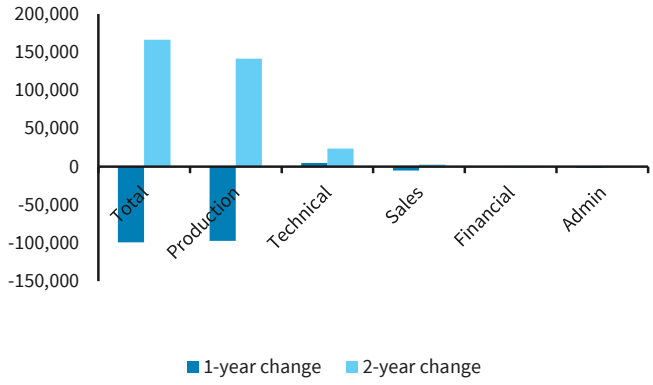
FIGURE 22. How AI is discussed by companies with the largest labour reduction in 2025

Company	Sector	AI adoption	Revenue impact	Cost impact	2025 employee changes	Labour impact rationale
BYD Company	Automobile	Manufacturing, battery production, robotics, intelligent driving, digital twins, AI chips and vehicle software.	No disclosure	40% lower battery defects, 20% longer battery life, and highly automated production.	-10%	No direct linking to AI
Baidu Inc.	Internet	Central to strategy across cloud, advertising, enterprise software, foundation models and autonomous driving.	AI-related revenue totalled CNY40bn in 2025	No disclosure	-7%	No direct linking to AI
ICBC	Bank	Digital-transformation strategy, operational processes, customer service, risk management and intelligent banking initiatives.	No disclosure	"Lingxing AI+" platform reportedly processed work equivalent to 55,000 person-years through more than 500 AI applications.	-1%	No direct linking to AI
Ping An	Insurance	Embedded across insurance, banking, healthcare, customer service, claims processing, fraud detection, and internal productivity systems.	AI agents supported CNY30bn of sales activity.	Contribution to key efficiency metrics	-16%	No direct linking to AI
Sinopharm	Healthcare	Transformation, logistics, pharmaceutical technology and innovation initiatives	No disclosure	No disclosure	-12%	No direct linking to AI
Hisense Home	Household durables	AI is becoming central to smart-home products, connected appliances, manufacturing systems and product innovation	A strategic driver but no revenue disclosure	No disclosure	-32%	No direct linking to AI
SAIC Auto	Automobile	AI is deployed in intelligent driving, autonomous vehicles, smart factories, mobility services, digital R&D and manufacturing.	No disclosure	No disclosure	-4%	No direct linking to AI
Tongwei	Electrical equipment	Operations, smart manufacturing, anomaly detection, predictive maintenance, quality management and process optimization.	No disclosure	Contribution to key efficiency metrics.	-20%	No direct linking to AI

Source: Company data, Barclays Research

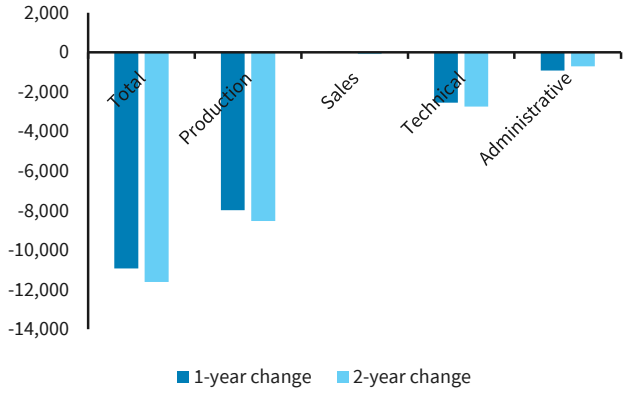
Production roles appear most exposed. At **BYD**, 98% of its ~99,000 workforce reduction came from production functions. **Tongwei** cut its workforce by 20% y/y in 2025, with 73% of the decline concentrated in production-related jobs. Neither company has formally attributed these reductions to AI deployment or automation.

FIGURE 23. BYD: Employee changes by function



Employee number
Source: Company data, Barclays Research

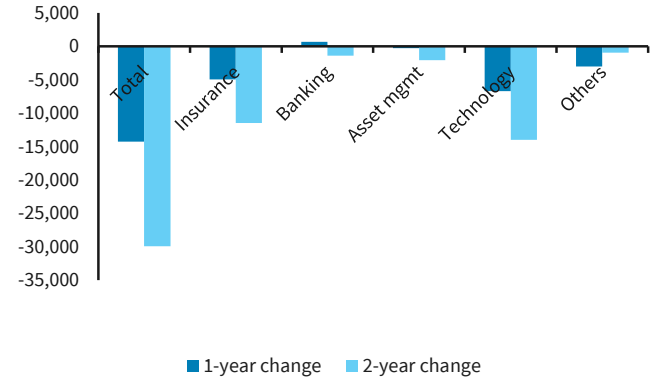
FIGURE 24. Tongwei: Employee changes by function



Employee number
Source: Company data, Barclays Research

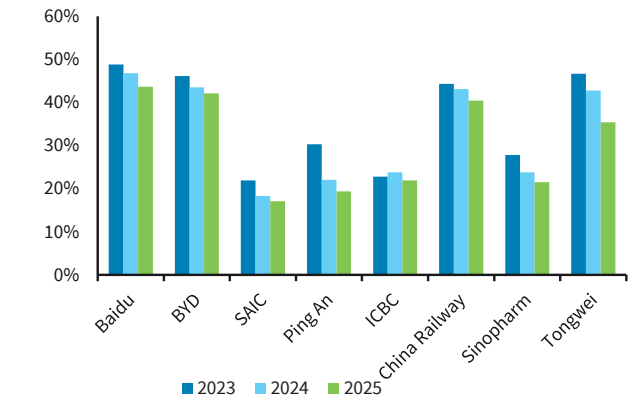
Outside manufacturing, **Ping An** reduced its workforce by 5%, with cuts primarily affecting insurance and technology roles. More broadly, our sample shows a consistent decline in employees aged under 30, pointing to potential pressure on entry-level and junior positions as companies pursue greater efficiency.

FIGURE 25. Ping An: Employee changes by function



Employee number
Source: Company data, Barclays Research

FIGURE 26. Younger employee (<30 years old) contribution has reduced across all key companies

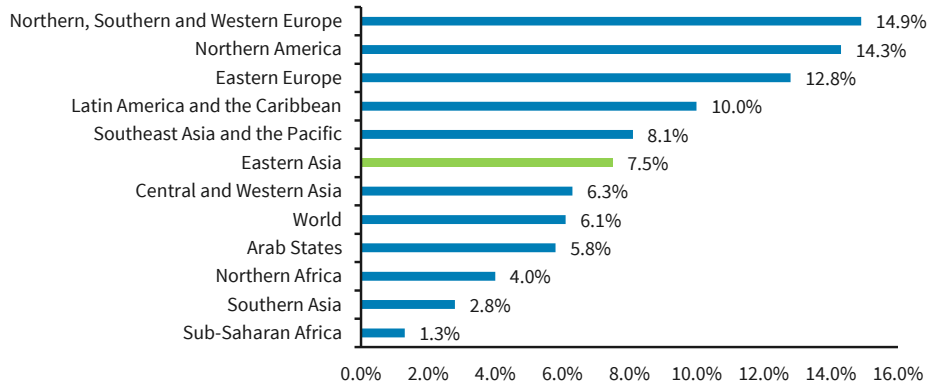


Source: Company data, Barclays Research

Although companies rarely discuss the age profile of workforce reductions, younger workers may be particularly vulnerable. The ILO estimates global youth unemployment reached 12.4% in 2025 and warned that AI could further pressure labour-market outcomes[16]. Around 6.1% of youth jobs worldwide are highly exposed to AI disruption, increasing to 7.5% in East Asia.

[16] "Global Employment Trends for Youth 2026", International Labour Organization, August 2026

FIGURE 27. **Exposure of youth job to AI disruption**



Source: International Labour Organization, Barclays Research

The next AI regulatory focus

Unlike the EU, China has not enacted a comprehensive AI law. Rather, it regulates AI through an evolving framework spanning cybersecurity, generative AI, algorithms, deep synthesis and content labelling, underpinned by technical security standards. The regime is primarily risk based, targeting applications with potential implications for health, behaviour, public opinion, safety and sustainable development.

A notable regulatory development came on 7 September 2026, when the Supreme People's Court published the Opinions on the Trial of Cases Involving Artificial Intelligence Disputes, marking China's first dedicated judicial guidance on AI. The document provides a legal framework for lower courts covering copyright, patent, data protection and contractual disputes, while also addressing emerging issues such as AI-generated likenesses and voice cloning. The move signals a growing focus on establishing legal accountability alongside AI innovation.

Two provisions stand out. Article 11 clarifies responsibility in **autonomous-driving accidents**, addressing manufacturer defects, driver fault, misleading automation claims and access to driving data. Article 13 sets out a framework for assessing **open-source AI liability**, based on licence conditions, rights restrictions, security controls and disclosure standards.

In our view, China is moving towards **greater AI accountability, not heavier regulation**. For investors, clear beneficiaries are companies capable of turning compliance infrastructure into customer trust and enterprise products. **Key watch points should cover content-heavy models, training pipelines and physical AI businesses demonstrating safety and event-level accountability.**

Thematic Investing: how adoption focus differs in the US

Hannah Greenberg BCI, US | William Thompson BCI, US

Our recent thematic analysis, Positioning for AI Disruption: The Next Phase of Enterprise Adoption, provides a useful comparison of how AI deployment is evolving in the US. As AI capabilities continue to advance rapidly, elevated investment across the value chain has increased scrutiny around ROI. Even as the cost per unit of intelligence declines, broader adoption and the shift toward more complex and agentic workflows is driving enterprise spend higher. In response, companies are re-evaluating how AI is deployed from both a technology and application perspective, with a greater emphasis on controlling the intelligence stack and proven execution.

Enterprise Spend Is Rising Despite Cost Declines

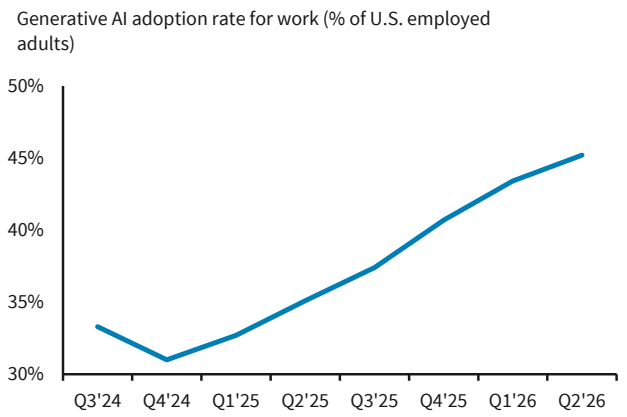
Since publishing Positioning for AI Disruption (US Edition & EU Edition) earlier this year, the AI cost paradox has become more apparent. While the upfront costs of AI deployment remain high, advances across hardware, the compute layer, and model efficiency are driving greater intelligence per watt. At the same time, the rise of open-weight models, including highly-capable Chinese models, has intensified competition. This has resulted in a broad spectrum of price-performance options, with increasingly capable low-cost models at one end and premium reasoning models at the other. As the performance gap has narrowed, open-weight models have started capturing a growing share of US inference demand. However, share of tokens does not translate directly to share of spend, given differences in pricing.

Despite ongoing declines in token pricing, enterprise spending continues to rise. Enterprise AI adoption is broadening across both users (Figure 28) and use cases, signaling deployment is maturing. At the same time, the shift toward agentic AI is underway. Data from OpenAI and Anthropicsuggests the work being done with AI is changing, with agentic use cases beginning to expand beyond coding work. Agentic systems use multi-step reasoning to break tasks into a series of actions. As agents plan, retrieve relevant context, call tools, analyse results, and retry when needed, each additional model call consumes tokens, driving higher token usage per task.

Insights from our sector analysts suggest AI adoption is starting to progress beyond the pilot phase, as companies begin to quantify the benefits and set strategic targets. In our view, this represents a key inflection point for the theme. Recent transcript analysis from our US Equity Strategy team revealed that although nearly half of S&P 500 companies discussed AI during Q2 2026 earnings, only 12% quantified AI-related value capture (Figure 29).

Taken together, we see US companies moving from a phase focused on AI access and experimentation toward one centred on execution, monetization, and measurable business outcomes.

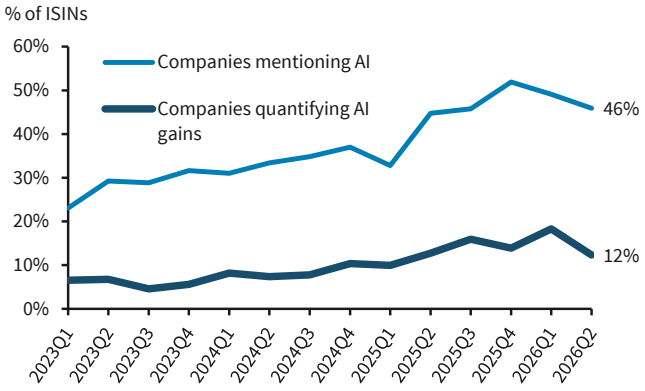
FIGURE 28. Generative AI adoption for work is rising in the US



Share of employed adults aged 18-64 who use generative artificial intelligence for work

Source: The Rapid Adoption of Generative AI, A. Bick, A. Blandin, and D. Deming, Management Science (2026) (https://pubsonline.informs.org/doi/10.1287/mnsc.2025.02523).

FIGURE 29. S&P 500 companies willing to quantify AI-related gains are a fraction of those discussing AI without quantification



Source: Barclays Research. For additional insights and the methodology, see Pulse check on AI adoption among S&P 500 companies and Positioning for AI Disruption: Insights from Q2'26 Earnings Calls.

Industry Response: Managing AI Spend

The industry is responding to rising AI spend with an emphasis on managing deployment costs while maximizing ROI. We see the response centred on two fronts:

- **Controlling the intelligence stack:** From a technical perspective, advances in AI frameworks, harnesses, and routing technologies are making it easier to separate applications from individual models. Rather than defaulting to the most capable and often most expensive model, enterprises can increasingly choose the model best suited to a particular use case based on performance, cost, latency, and security considerations. Model routing systems could help reduce switching costs and make open-weight models easier to deploy.

- **Strategic focus on ROI:** From an organizational perspective, companies are moving away from broad experimentation toward more targeted deployment strategies. Organizations are placing greater emphasis on ROI, workflow redesign, and governance. Importantly, ROI discipline does not necessarily imply lower AI spending. As AI unlocks new revenue opportunities and cost savings, companies may reinvest in further deployment. In addition, lower AI costs expand the number of use cases capable of generating attractive returns.

Implications Across the Value Chain

As the focus shifts from adoption to value realization, execution capabilities are becoming the key differentiator. As access to highly-capable, lower-cost models becomes more widespread, model access alone is becoming less of a source of competitive advantage. Instead, competitive positioning increasingly depends on an organization's ability to deploy AI effectively and translate adoption into measurable business outcomes. In our view, this elevates the importance of company-specific factors, such as proprietary data, technology readiness, and AI talent.

Within our analysis, we identified **30 US companies** demonstrating AI implementation progress, reporting quantifiable benefits, or possessing company-specific characteristics that may support successful execution. These span Financials, Healthcare, Consumer, Industrials, and Energy sectors. In parallel, **software vendors are playing a critical role in operationalizing AI adoption**. While foundation models are becoming increasingly intelligent, they do not

inherently understand a company's business context, workflows, and proprietary data. Software vendors bridge this gap by connecting AI intelligence to enterprise workflows, data, governance, and systems of record. In our view, this creates opportunities across application software, infrastructure software, cybersecurity, and vertical SaaS. For company-level details, see Positioning for AI Disruption: The Next Phase of Enterprise Adoption.

US Potentially Faces Converging Data Center Supply Constraints

As AI adoption continues to accelerate, we believe supply-side constraints remain the key determinant of the pace of AI infrastructure deployment in the US. As outlined throughout our Powering AI series, the industry faces a convergence of execution risks across power, capital, and skilled labor, alongside increasing local opposition, permitting delays, and growing regulatory fragmentation around data centre development.

These challenges stand in sharp contrast to China, where power availability and permitting are less likely to represent binding constraints on AI infrastructure build-out. The DeepSeek moment in early 2025 was widely viewed as AI's "Sputnik moment," raising concerns around democratic AI leadership, China's technological ambitions, and the ability of the US to deploy sufficient infrastructure to maintain its competitive position (Deep impact: The global AI arms race). The result has been a growing focus on sovereign AI capabilities, increasing urgency around digital infrastructure deployment, domestic semiconductor manufacturing, AI hardware export controls, and access to reliable, affordable energy.

While policymakers increasingly view AI leadership and the infrastructure that supports it as strategic economic and national security priorities, many of the decisions that ultimately determine whether, where, and how quickly data centres are built remain in the hands of state and local governments. At the same time, communities are becoming increasingly focused on electricity costs, water and land use, emissions, and other local impacts associated with large-scale data center development, as we detailed in Powering AI: The Politics of Data Centers.

Water consumption has become a particularly contentious issue, despite widespread misconceptions surrounding data center cooling. As we discussed in Powering AI: Demystifying Data Center Cooling & Water Use, lower water consumption is not always better given the multidimensional tradeoffs between water use and power consumption. Efforts to reduce on-site water use associated with evaporative cooling can increase electricity demand, potentially resulting in higher power-related emissions and greater indirect water consumption at thermal power plants.

In some regions, concerns about AI itself, including job displacement, security risks, and a lack of trust in large technology companies, are beginning to emerge alongside more traditional environmental and infrastructure concerns. This is creating growing tension between national AI ambitions and the interests of the communities expected to host the infrastructure.

International expansion is not always a viable solution. Moving AI infrastructure into jurisdictions such as Canada, Mexico, or other foreign markets introduces questions around data sovereignty, national security, regulatory oversight, and legal jurisdiction. Governments and enterprises increasingly want sensitive data, AI models, and critical digital infrastructure to remain under domestic control, particularly as AI becomes more strategically important. Data residency, cybersecurity, privacy, and sovereign AI initiatives may therefore limit the extent to which AI workloads can simply be relocated across borders in pursuit of lower costs, easier permitting, or faster access to power.

Credit research: Commercialisation and funding

Roanna Chau[(i)] Barclays Bank, Hong Kong

AI adoption is becoming more commercially visible

China TMT companies, spanning internet, platform and hardware, are beginning to show clearer commercial benefits from AI, although the evidence remains uneven across companies (Figure 30). This to a certain extent is consistent with the broader adoption indicators discussed in Understanding China's AI roadmap, which show adoption led by manufacturing and ICT but remaining uneven across sectos and company sizes.

The most directly measurable revenue is concentrated among cloud and infrastructure providers. Alibaba, Baidu and Tencent, for instance, are seeing stronger demand for AI compute and cloud services, while Kuaishou has established a visible application-level revenue stream through Kling AI. Lenovo provides the clearest hardware read-through, with AI-related revenue spanning devices, servers, infrastructure and services.

Elsewhere, commercialisation appears more embedded and is less directly measurable among the application-led companies. At JD.com, Meituan, Tencent Music Entertainment and Weibo, AI is primarily improving recommendation, conversion, customer engagement, advertising performance and internal efficiency. JD.com is using AI across advertising, recommendation and supply-chain operations; Meituan across merchant tools, quick commerce and internal productivity; and Tencent Music Entertainment and Weibo to enhance content discovery, engagement and advertising performance. These use cases are consistent with the productivity-led adoption discussed in the FX and Rates: Stronger external support, limited domestic reflation section, where AI is being embedded across financial services, healthcare, logistics, media and entertainment, often through workflow efficiencies rather than separately reported AI revenue.

In other hardware and component names. Xiaomi has begun generating revenue from API and token services, but AI remains embedded within a broader range of product offerings spanning smartphones, AIoT and EVs. AAC Technologies, meanwhile, is positioned as a hardware and/or components supplier through liquid cooling, thermal management, optical components, edge-AI devices and robotics rather than as a developer of proprietary AI models. Lenovo, Xiaomi and AAC Technologies' product offerings are well positioned to capture China's near-term export and manufacturing-led AI growth (see Macroeconomics: Gauging AI implications for more details).

FIGURE 30. Reported commercial evidence of AI adoption

Issuer	Commercial evidence
Alibaba (BABA)	AI Cloud and Compute revenue +44.9%; AI products 35% of external cloud revenue; AI-product annualised revenue approaching USD10bn
Baidu (BIDU)	AI Cloud Infrastructure +50%; GPU cloud +283%; AI-powered businesses are an increasingly material part of Baidu Core
Kuaishou (KUAISH)	Kling AI revenue above CNY850mn, +200% y/y
Lenovo (LENOVO)	AI-related revenue USD9.3bn, +60%, or 35% of group revenue
Tencent (TENCNT)	Cloud growth accelerated to low-20%; improved AI-enabled advertising targeting and commercial cloud use cases

Source: Company data, Barclays Research

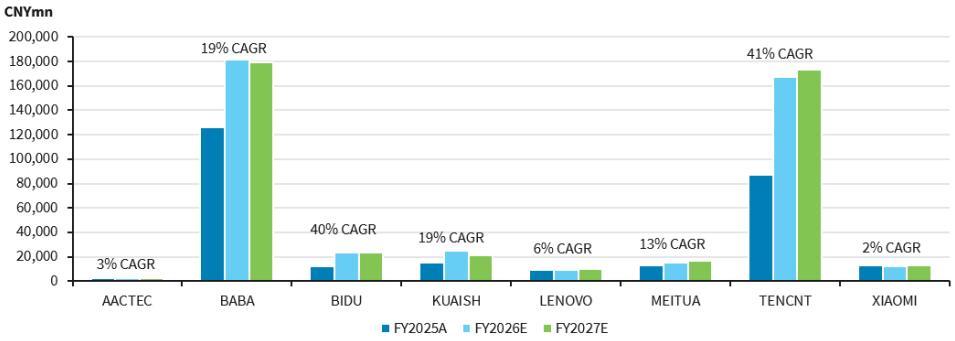
Spending continues to accelerate

China TMT's AI investment cycle is broadening, but the capex acceleration remains heavily concentrated in the infrastructure-led platforms, rather than sector-wide. At the economy-wide level, this forms part of the broader expansion in compute capacity described earlier in

Understanding China's AI roadmap, including the potential CNY2trn national data centre programme and the expected buildout of data centre capacity through 2030.

Alibaba and Tencent are already operating at a different scale from the rest of the sector. Their 2Q26/Jun-Q capex totalled approximately CNY120bn, driven by data centres, servers and chips, model training, inference capacity and cloud infrastructure. Alibaba had spent CNY190bn of its CNY380bn three-year plan by end-June, while Tencent's operating capex increased 190% y/y, to CNY51.8bn. Baidu is ramping investment more aggressively from a smaller base, with quarterly capex almost doubling sequentially, to CNY11.4bn, in 2Q26. Kuaishou also stepped up compute investment for Kling AI, although the company indicated that spending was front-loaded in 1H26.

FIGURE 31. Consensus capital expenditure forecasts



BABA and LENOVO fiscal year ends March; Capex CAGR shown at company level for 2025-2028
Source: Bloomberg, Barclays Research

FIGURE 32. 2Q/Jun-Q capex spending and company guidance

Issuer	2Q/June-Q Capex (CNYbn)	Capex guidance	Execution/progress	AI-related investment(s)
AAC Technologies (AACTEC)	1.3*	n.a.	n.a.	AI infrastructure componenets, Edge AI hardware, embodied AI etc.
Alibaba (BABA)	67.7	3-year CNY380bn (announced Feb 25)	CNY190bn by end-June 26	Full-stack spending spans data centres, compute, proprietary T-Head GPU/CPU/networking chips, Qwen models and AI applications etc.
Baidu (BIDU)	11.4	n.a.	n.a.	AI Cloud infrastructure, GPU cloud, Kunlunxin chips, ERNIE, AI Search, digital humans, agents and Apollo Go etc.
Kuaishou (KUAISH)	8.7^	n.a.	n.a.	Kling AI, recommendation models and internal productivity tools etc.
Lenovo (LENOVO)	1.4	At least USD1.4bn annually through FY27–28	n.a.	Qira and Tianxi personal-AI agents, hybrid enterprise AI, AI servers, liquid cooling and AI infrastructure etc.
Meituan (MEITUA)	3.2^	n.a.	n.a.	LLMs, AI at work and AI in products, including LongCat-2.0, quick-commerce tools, merchant applications and Xiaotuan etc.
Tencent (TENCNT)	52.8	Low-teens % of revenue	n.a.	Hunyuan model training, WorkBuddy and CodeBuddy inference, Weixin AI initiatives and cloud services etc.
Xiaomi (XIAOMI)	3.6	CNY16bn for FY26	CNY6.9bn by end-June 26	Smartphone, AIoT, Smart EV, AI etc.

Source: Company data, Bloomberg, Barclays Research

Who can fund the buildout, and how?

Is AI growth keeping pace with spending? Not yet, at least not consistently, although the growth trajectory is improving. Commercial progress is narrowing the gap, but most issuers remain in the buildout phase. Therefore, the key near-term question is not only whether companies can fund this timing gap, but also how they choose to do so.

Issuers with stronger recurring cash generation may be able to self-fund the investment cycle for longer, giving them greater flexibility to pace external borrowing Where additional capital is required, we expect opportunistic and diversified funding activity rather than a broad wave of bond issuance. Potential channels include domestic bank financing, bond issuance (USD and CNH), convertible bonds, and share placements. Companies may also preserve liquidity by reducing share repurchases.

Unlike some US hyperscalers, we do not expect China TMT issuers to establish a once-a-year bond-issuance cadence to fund AI investment. However, issuers with substantial investment ambitions may still access the market regularly when conditions are attractive or when they wish to pre-fund future spending. The pace of supply will therefore depend not only on immediate liquidity needs, but also on investment opportunities, market access, refinancing strategy and issuers' desired liquidity buffers. The liquidity walks that we have carried out assess the scale and timing of potential shortfalls (see full report China TMT: AI adoption meets the funding test for details). Our analysis identifies Baidu as the issuer most likely to require additional funding in the next 15 months, followed by Alibaba later in the forecast horizon. Kuaishou remains adequately funded in the base case, although its cushion is relatively thin. AAC Technologies, Lenovo, Meituan, Tencent and Xiaomi retain greater liquidity headroom, allowing them to pace borrowing more flexibly. Any issuance from these names may support refinancing, pre-funding, additional investment or liquidity preservation, rather than indicate an immediate cash shortfall.

However, **the funding question could become more complex as compute demand expands and as Chinese TMT companies extend beyond domestic procurement into overseas data centres and GPU capacity**. Rather than funding all infrastructure directly, issuers may increasingly act as tenants, customers or offtakers, with data centre operators and infrastructure partners or project companies undertaking the investment and raising the associated debt.

Moody's estimates that APAC data centre expansion will require approximately USD900bn of investment: USD500-550bn for servers and computing equipment; USD200-250bn for land, buildings, fibre and supporting infrastructure; and USD90-110bn for additional power generation. Funding could be provided through cloud leases, project companies, joint ventures, third-party operators or customer-backed arrangements.

The financing mix may vary by asset and development stage. Land acquisition and early development are typically funded through sponsor equity, private equity, infrastructure funds, local partners or tenant joint ventures. Construction can introduce bank loans, project finance, bilateral facilities, sustainability-linked debt and project bonds. Once utilisation and contracted cash flows are established, mature assets may access bank term loans, infrastructure trusts or REITs, ABS or CMBS, private credit and corporate bonds.

This could also broaden the investible China AI credit universe beyond existing TMT issuers. Data centre operators, cloud providers and project vehicles may emerge as borrowers in the bond market, with Chinese TMT companies supporting their credit profiles through long-term leases, capacity reservations or offtake commitments, or potentially providing some sort of guarantee.

At the moment, Alibaba Cloud reports a presence across 31 regions, including markets across Southeast Asia, Northeast Asia, the US, Latin America, Europe and the Middle East, and has outlined an ambition to operate more than 20GW of global data centre capacity by 2032. Tencent Cloud is present across 25 regions, including the US, Europe, Asia, Latin America and the Middle East, and has announced plans for its first Saudi Arabian data centre. Outside Alibaba and Tencent, quantified overseas commitments are less common and generally relate to market-entry or manufacturing plans rather than cloud and data centre infrastructure.

Macroeconomics: Gauging AI implications

Yingke Zhou Barclays Bank, Hong Kong | Ying Zhang Barclays Bank, Hong Kong

Near term: China's AI growth boost comes mainly from exports not investment

We believe AI's growth dividend for China comes primarily through exports not investment. We estimate AI could add around 0.8ppt to GDP growth in 2026 and 0.5ppt in 2027. While smaller than the roughly 1ppt boost in the US in 2026, the contribution is still meaningful. Without AI-related investment and export demand, China's growth rate would likely fall below 4%.

- **Investment is the first transmission channel, but its macro impact is relatively modest.** We estimate that China's AI-related investment will likely reach CNY960bn in 2026 and hit CNY1.25tn in 2027 (Figure 33), reflecting rising spending by hyperscalers, telecom operators and data-centre developers. AI is generating new investments spanning data centres, cloud infrastructure, semiconductors, networking equipment and power systems.

  However, the macro impact of AI investment remains modest. We estimate that AI-related investment accounts for less than 2% of China's total investment, making it too small to fully offset the drag from property and slower infrastructure spending (these two combined accounted for 67% of total investment). After adjusting for imported chips and equipment, and land that do not directly add to current real GDP, we estimate AI-related investment could contribute roughly 0.3ppt to annual GDP growth in 2026 followed by 0.2pp in 2027.

  Chinese hyperscalers are not necessarily investing less (than their US counterparts, Figure 34) because they are less enthusiastic about AI. Rather, investment decisions are also being shaped by factors such as AI monetization, cash flow dynamics, and high-end chip constraints. China's AI monetization remains far behind the US. According to the Rhodium Group, all of China's AI models combined generate only about 10% of the revenue reported by OpenAI and Anthropic[17]. Moreover, Chinese hyperscalers' cash flow is not keeping pace with capex. Chinese hyperscalers are effectively using profits from their legacy, consumption-linked businesses to subsidize AI investment, but weaker household demand suggests this source of funding is under pressure.

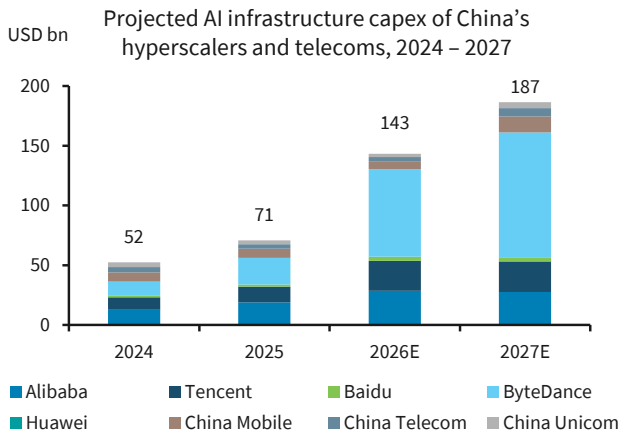
- **Exports are the more important growth channel.** China is emerging as the manufacturing hub of the global AI boom, supplying servers and data-processing equipment, AI-enabled consumer electronics, semiconductors, cooling systems, energy storage system, and other AI-related hardware. In 2026, China's AI-related exports grew 52% y/y YTD, accounting for 23% of total exports (Figure 35 and 4). These products contributed nearly 10pp to overall export growth, accounting for 51% of total export growth, making them the single largest driver of China's export expansion in the first eight months of 2026.

  The export boost is more meaningful than the investment boost. As global AI spending accelerates, China's manufacturing scale and supply-chain advantages position it to capture

[17] see "Examining China's AI Financing", Rhodium Group, September 2026

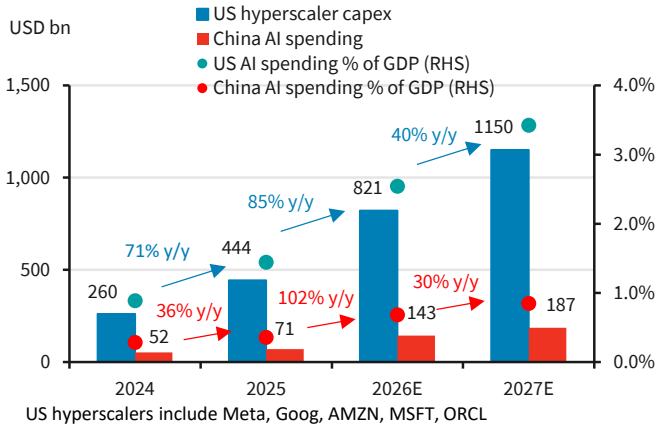
a disproportionate share of the hardware cycle. We estimate AI-related exports could add around 0.5ppt to GDP growth in 2026, moderating to 0.3ppt in 2027 as the AI capex boom among US hyperscalers gradually normalizes. In short, the near-term AI story for China is less about productivity and more about exports and investment, with exports likely contributing the larger share of growth.

FIGURE 33. AI capex in China is expected to see a visible pickup in 2026...



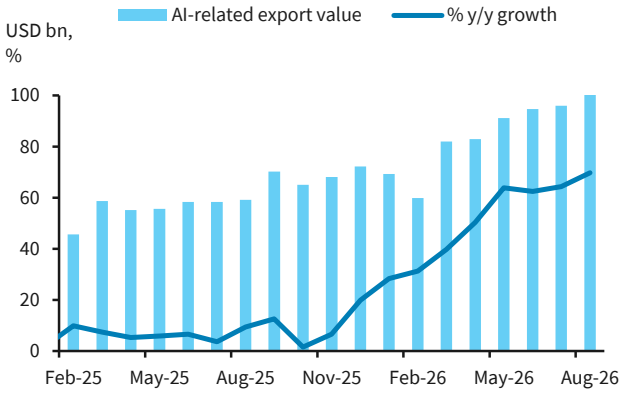
Source: Bloomberg, Barclays Research

FIGURE 34. ...but the scale remains relatively small compared with the US



Source: Bloomberg, Wind, Barclays Research

FIGURE 35. AI-related exports increased sharply...



Source: Wind, Barclays Research

FIGURE 36. ...accounting for ~1/4 of China's total exports now



Source: Wind, Barclays Research

Longer term: Putting AI into potential growth model

In the longer term, the bigger opportunity from AI lies in productivity growth rather than cyclical demand support. Its main contribution comes through higher total factor productivity (TFP), as firms use AI to automate routine tasks and improve efficiency across manufacturing and services. The IMF estimates that AI could lift global potential growth by as much as 1pp over five years, equivalent to an average annual boost of 0.2pp, with the gains increasing as generative AI adoption becomes more widespread.

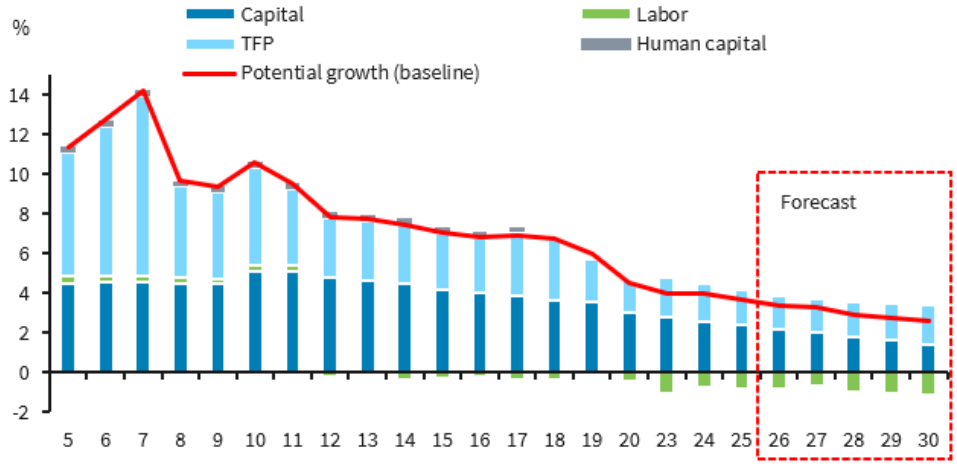
However, while AI may slow the decline in potential growth, it is unlikely to fully reverse it. To gauge the impact, we rely on a standard Cobb-Douglas production function framework:

$Y(t)=A(t)*K(t)^{\alpha}*(L(t)H(t))^{1-\alpha}$.

Where: Y = real GDP, A = total factor productivity (TFP), K = capital stock, L = number of employees, H = human capital, α = elasticity of output to capital, 1-α = elasticity of output to labour, and t = years.

With a faster-shrinking and aging population, and diminishing returns to capital accumulation, we expect China's potential growth to continue trending lower despite AI-driven productivity gains. In our baseline scenario, AI, automation and machine learning provide some measured boost to TFP growth, partially offsetting demographic and investment headwinds. Even so, we expect potential growth to slow to approximately 2.5% by 2030 from 4% in 2021-25 and from above 10% in 2005-07. This trajectory is broadly consistent with forecasts generated by researchers at the ADB[18], IMF[19] and NY Fed[20].

FIGURE 37. China's potential growth and inputs contribution breakdown



Source: IMF, Wind, Barclays Research

AI productivity gains in practice: Evidence across sectors

In practice, AI is delivering measurable productivity gains across major industries.

- **In financial services**, AI is replacing large volumes of routine work. ICBC's AI digital employees perform work equivalent to more than 55,000 person-years, while China Merchants Bank reported 15.6mn hours of labor substitution in 2025 under its "Human + Agent" model.
- **In healthcare**, AI is helping deliver more services with fewer resources. JD Health's AI family doctor recorded over 30 million consultations within six months, while BGI has significantly reduced genome-analysis time through AI-powered automation and diagnostics.
- **In logistics**, AI is replacing labor-intensive coordination with algorithm-driven decision-making. SF Express uses AI for scheduling and autonomous delivery, Cainiao operates a "logistics brain" for network optimization, and Didi relies on AI to improve real-time dispatch and fleet utilization.

[18] "ADB East Asia Working Paper Series: The Long-Term Growth Prospects of the People's Republic of China," Peschel and Liu, ADB, 2022. *These authors forecast China's growth averaging 5.3% in 2020-25, 3.5% in 2026-2030, 2.7% in 2031-35, and 2% in 2036-2040.*

[19] "People's Republic of China: Selected Issues: Sustainable and Balanced Growth in the Longer Term," IMF Asia and Pacific Dept, 10 Feb 2023. *In this analysis GDP growth estimates fall to an average of ~4% in 2023-27 and ~3% for 2028-37.*

[20] "China's Growth Outlook: Is High-Income Status in Reach?" Higgins, Federal Reserve Bank of New York, Economic Policy Review 26, no. 4, October 2020. *The 'Pretty Good' scenario in this analysis saw growth per capita average 3.8% for 2018-28 and 2.1% in the decade to 2038.*

- **In media and entertainment**, AI is accelerating content creation and distribution. Douyin's recommendation engine can generate targeted content suggestions within seconds of app launch, while miHoYo is integrating AI-powered NPCs and production tools across the game-development process.

AI is not creating (enough) jobs yet in China: evidence from three sectors

As discussed in Global Outlook: Reluctantly optimistic, US AI boom has created a meaningful employment impulse in the US because it is layered on top of an already large construction sector with more than 8 million workers. Data centres are driving demand not only for software engineers, but also for electricians, plumbers, pipefitters, steel workers, contractors, telecom engineers, and power infrastructure workers. Despite around 20% contraction in property construction in 2025 vs 2021 in the US, the construction employment actually rose by 10% to 8.3mn in the US market, accounting for 5% of total nonfarm employment.

But China is different. We do not think AI is creating meaningful employment in China yet.

- **Construction.** AI investment does create jobs in data centres, power equipment, servers, and industrial automation, but the scale is too small. Unlike in the US, where the AI buildout is layered on top of a large construction sector, China's data-center expansion is highly industrialized, relying heavily on prefabricated components, factory-built modules, and standardized designs that compress construction timelines and reduce labor requirements. A notable example is Alibaba's"100-Day Datacenter" model, which combines 30 days of factory prefabrication and site preparation, 50 days of installation, and 20 days of commissioning, taking just 100 days from a completed shell to server deployment[21].

  More importantly, AI-related construction activity remains too small to offset the much larger employment drag from the property downturn. The prolonged adjustment in the property market has reduced demand for construction workers significantly, especially migrant labor, across real estate development, infrastructure, building materials, furniture, and related supply chains. As a result, the number of construction workers in China was roughly 25% lower in 2025 than at its 2021 peak (Figure 38), following the property bubble burst. In summary, AI-related construction activity can provide a modest offset, but it is nowhere near large enough to replace the labor demand once generated by the property sector.

- **Manufacturing.** AI is not likely to become a source of net job creation in manufacturing either. The surge in AI-related exports has been largely a price and value-added story rather than a volume story, and these industries are far more capital-intensive than labor-intensive. Moreover, Chinese firms are adopting AI, robotics, and automation primarily to improve efficiency, and reduce labor intensity. AI enables factories to produce more with fewer workers, particularly in routine assembly, quality control, logistics, and administrative functions. As a result, the number of manufacturing workers in China in 2025 was roughly 30% below its 2013 peak. In other words, AI is boosting productivity and output, but not generating enough new jobs to offset the broader decline in manufacturing employment.

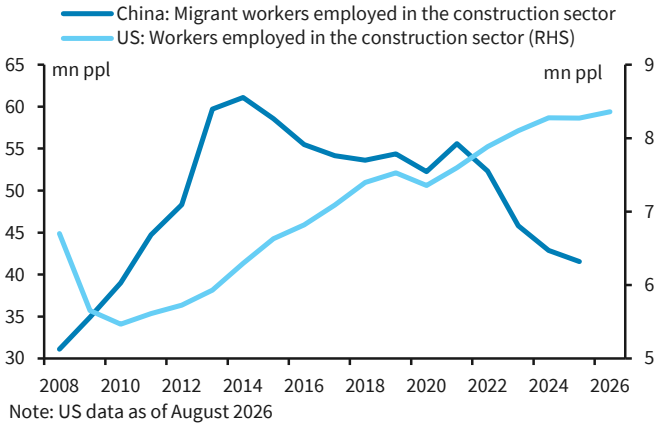
- **Services.** The same is true in services. Most corporate AI adoption is concentrated in customer service, back-office operations, coding assistance, financial processing, marketing, and document review, where the primary goal is to automate routine tasks and reduce headcount, not expand payrolls. While AI is creating some high-skilled jobs in software engineering, data science, model development, and cloud infrastructure, the numbers are too small to offset broader weakness in white-collar hiring. For context, China's entire IT sector, the industry most directly linked to AI, employs only around 8 million workers, less

[21] see "Alibaba says modular design delivers AI data centres in 100 days at 10% lower cost", SCMP, 11 August 2026

than 1.5% of total employment. The employment base is simply too small to move the needle for the overall labor market.

More importantly, AI has yet to become a major job creator, even within the AI ecosystem itself. The combined workforce of Alibaba, Baidu, ByteDance, Huawei, Tencent, China Mobile, China Telecom, and China Unicom declined by 6.6% in 2026 from its recent peak in 2022 (Figure 39). While some of the decline reflects corporate restructuring, such as Alibaba's asset divestitures, the broader trend underscores a key point: AI is primarily a productivity story, not an employment story.

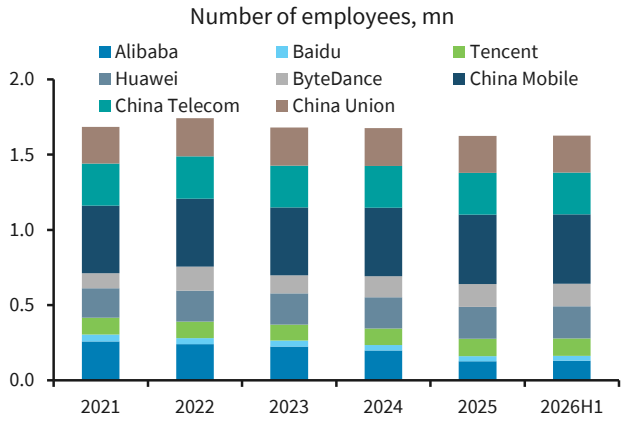
FIGURE 38. Unlike the US, AI-driven construction is far too small to offset property-related job losses



Note: US data as of August 2026

Source: Wind, Barclays Research

FIGURE 39. AI is not proving to be a major job creator even within the AI ecosystem, yet



Source: Company annual reports, Bloomberg, Wind, Barclays Research

China's AI inflationary buildout is behind us; disinflation likely lies ahead

Conceptually, AI follows a two-stage inflation cycle: inflationary when you build it, disinflationary when you use it. The inflationary phase is driven by a massive investment boom in data centres, power generation, grid transmission, semiconductors, and related infrastructure. When demand for capital, energy, and skilled labor rises faster than supply can respond, prices move higher. This dynamic is still evident in the US, where power and grid constraints have become key bottlenecks to AI deployment. China, by contrast, has already spent years building out power, transmission networks, industrial capacity, and digital infrastructure. As a result, the most inflationary part of the AI infrastructure-related buildout is likely behind China (see China AI Watch: More growth, less reflation, 8 October 2026).

FX and Rates: Stronger external support, limited domestic reflation

Mitul Kotecha[i] Barclays Bank, Singapore

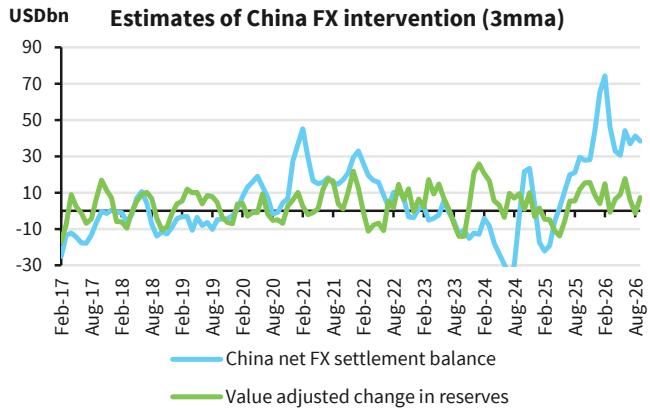
FX: AI adds to the CNY's external anchor

AI-related exports now represent a significant share of China's exports. This should generate additional foreign-currency receipts and reinforce the structural support to the CNY from China's large trade surplus. However, the CNY response is unlikely to be proportional to the export boost. The conversion of export receipts into CNY will remain important. Moreover, the AI export cycle remains closely linked to foreign demand and capital expenditure by overseas hyperscalers. As the global AI capital-expenditure cycle eventually normalises, the boost to China's exports and growth is also likely to moderate.

There is also a policy constraint on appreciation. With domestic demand remaining weak and exports accounting for most of the AI growth dividend, policymakers have an incentive to preserve external competitiveness. A sharp rise in the CNY would tighten financial conditions and weaken the channel through which AI is currently supporting growth. The PBoC is therefore likely to continue managing the pace of appreciation, even if stronger AI exports reinforce appreciation pressure over time.

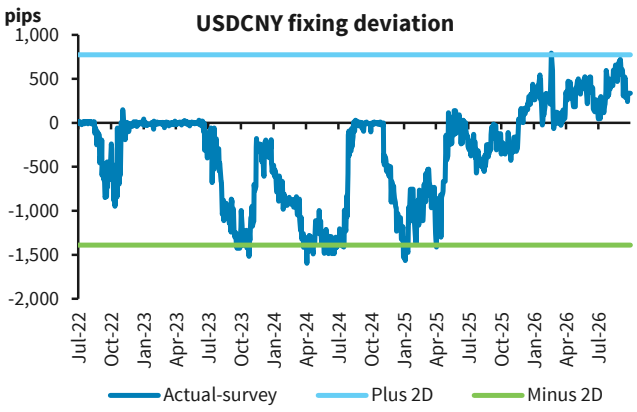
The net effect is an improvement in the CNY's underlying fundamentals rather than a discrete AI-driven currency rerating. We continue to expect USDCNY to grind lower, but gradually. AI-related export receipts should support the CNY and reduce depreciation risks, while weak domestic demand, China's low-yield environment and the PBoC's preference for an orderly adjustment should limit the pace of appreciation. The benefit may also be more visible in the CNY's trade-weighted performance than in the bilateral rate against a still-firm USD.

FIGURE 40. **Exporter conversions have slowed but remain elevated**



Source: Bloomberg, Barclays Research

FIGURE 41. **PBoC continues to manage the pace of appreciation via USDCNY fixings**



Source: Bloomberg, Barclays Research

Rates: AI supports CGBs more than CNY NDIRS

China's AI-driven growth impulse does not materially challenge the low-yield environment. AI-related capital expenditure remains relatively small compared with total fixed investment, while the associated gains have yet to generate meaningful employment, wage or consumption growth. Stronger AI-related output should eventually reduce the urgency for additional monetary easing without creating a case for PBoC tightening.

This combination remains supportive of CGB duration. Subdued underlying inflation, weak private credit demand and the continued property downturn should sustain duration demand from banks and other domestic institutional investors. AI-related export strength is unlikely to generate sufficient domestic reflation to drive a sustained rise in government bond yields. Moreover, the relatively modest scale of the domestic AI investment impulse suggests limited additional sovereign financing pressure.

The implications are somewhat less bullish for CNY NDIRS. The export and growth boost from AI reduces expectations of further PBoC rate cuts, while targeted financing for technology investment and reduced receiving demand could keep swap rates relatively sticky. NDIRS may also remain more sensitive than CGBs to global rates volatility, paying flows and changes in medium-term policy expectations. Stronger AI-related activity could therefore widen the

divergence between a CGB market supported by domestic institutional demand and an NDIRS market pricing less policy easing.

Within NDIRS, the front end should remain anchored while the 5y sector is more exposed to any improvement in growth, reduced easing expectations and any reassessment of China's medium-term potential growth. This supports a gradual 2y5y steepening bias. We continue to favour CGB duration while maintaining a 2y5y CNY NDIRS steepener recommendation, with the front end anchored by the PBoC and the 5y sector more exposed to reduced easing expectations and the AI-related growth impulse.

AI sharpens FX differentiation

The key FX divide is between economies integrated into the upstream AI supply chain and those competing with increasingly efficient Chinese downstream exports. In Asia, this should favour TWD and KRW and, to a lesser extent, MYR and SGD. However, the support may be more visible in relative-value Asian crosses than against the USD, given the continued influence of US yields and broad dollar conditions.

Taiwan and Korea are the clearest regional beneficiaries, as stronger demand for advanced semiconductors and memory supports exports, current-account surpluses and underlying appreciation pressure on TWD and KRW. However, the spot response may lag the trade impulse if exporter conversions slow or residents recycle export income into overseas assets. MYR and SGD should also benefit from complementary electronics production, data-centre investment and related services, with the effect transmitted through stronger external balances and investment inflows.

THB and PHP appear to be the clearest relative losers, given their more limited AI-export offsets and weaker external balances, while INR may lag because its exposure is concentrated in services rather than AI hardware. Outside Asia, AUD is the clearest commodity beneficiary, while CLP and PEN offer more conditional exposure to stronger copper demand. MXN also stands out as an underappreciated manufacturing winner through its exposure to US demand for electrical and power equipment.

Appendix

FIGURE 42. China's AI+ for industry and consumption priorities have broad economy coverage

Priority	Sub-group	Sectors	Targets
Industry	Manufacturing	Steel, petrochem, new materials, non-ferrous metals, building materials	
		Industrial mother machines, automobiles, power equipment, shipbuilding, aerospace.	
		Textiles/apparel, home/furnishing, food processing, pharmaceuticals, biomanufacturing, historical/classic industries.	
		Electronic components, consumer electronics, new displays, quality control, green/low-carbon electronics.	
		Software lifecycle toolchains, traditional software upgrading, vertical agents for medical/education/ finance/legal sectors, and software datasets.	3–5 general large models deeply applied in manufacturing, 1,000 industrial agents, 100 industrial high-quality datasets, 500 typical scenarios, 2–3 globally influential ecosystem-leading firms, and 1,000 benchmark enterprises
	Energy	Power grids, new energy business models, renewables, hydropower, thermal power, nuclear power, coal, and oil & gas	5+ professional large models in power grids, power generation, coal, and oil/gas, 10+ replicable demonstration projects, 100 typical application scenarios, and 100 technical standards.
	ICT	5G-A/6G, next-generation optical networks, IPv6+, industrial internet, network agents, intelligent network equipment, edge inference, compute networks, AI phones/PCs, smart home devices, wearables, information consumption, and vertical industries such as raw materials, electronic information, and equipment manufacturing	30+ high-value scenarios by 2028 and ≥75% coverage of 1-millisecond urban compute latency circles
	Transport	Combined assisted driving, intelligent rail, smart shipping, smart civil aviation, smart postal delivery, intelligent construction/maintenance, and intermodal transport/smart logistics	
	Agriculture	Agriculture data platforms, agriculture land maps, agricultural foundation models, smart farms, smart ranches, smart fisheries, full-chain agricultural digitization, and smart machinery	Agricultural production informatization above 30% by 2026 and 32% by 2028, plus 20+ foundational model algorithms/tools or SaaS services by 2028
Consumption	Goods	AI phones, AI computers, smart TVs, smart home appliances, smart kitchen/bathroom, smart lighting, AI glasses, wearables, intelligent connected vehicles, AI vehicle terminals, BCI, AR, VR, and MR products	
	Robots	Humanoid, quadruped, and bionic robots; elderly-care robots, companion robots, AI life assistants, health monitoring, mobility assistance, and smart housework	
	Services	Home services, elderly care, cultural tourism, hotels, catering, and education services	
	Commercial	Smart retail, smart stores, smart business districts, wholesale markets, e-commerce operations, customer service, design, marketing, livestreaming, smart logistics parks, smart ports, smart warehouses, rural logistics, pickup lockers, smart parcel boxes, low-speed unmanned delivery vehicles, and drones	

Source: Various government agencies, Barclays research

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